

第 54 屆全國技能競賽 青少年組



商務軟體設計職類 競賽試題及評分彙總表

競賽及評分項目

☒ 商務軟體設計

選手編號：_____

抽籤崗位號碼：_____

選手姓名：_____

總分：_____

評分裁判：_____

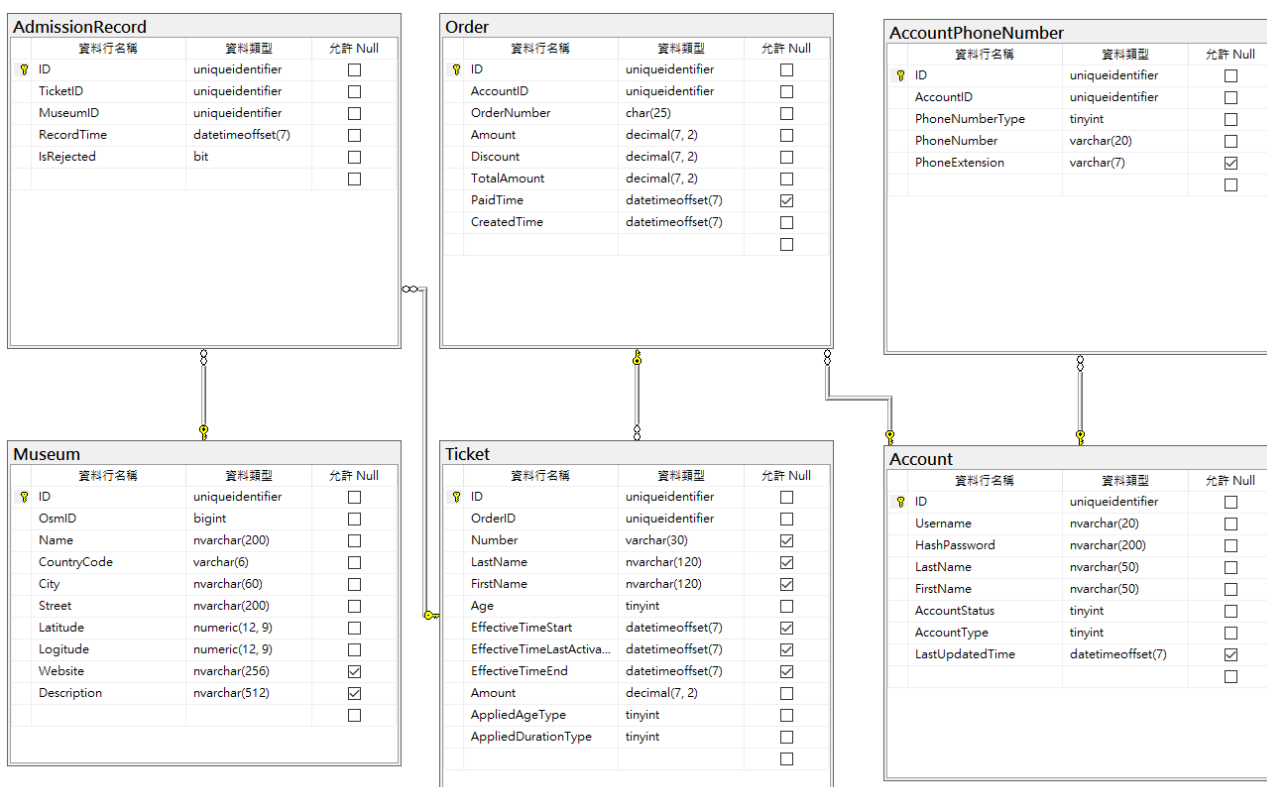
Scenario description:

The World Museum League has assigned you as a system analyst and developer to design the "World Museum League Pass System" management portal using development tools such as Visual Studio and SQL Server. Some system functions require the expansion of SQL database tables and data fields.

Section 1: Database Relationship

Your project manager has already conducted system analysis and preliminary SQL database design. The backup database file is named **WML_ANM.bak**. You must use the SQL database provided for this subject to continue refining the system functionalities. Additionally, some system functionalities will require the expansion of SQL database tables and data fields.

1. Please rename the SQL backup database **WML_ANM.bak**, replacing the last two characters "NM" with your position number.
2. The database relationship diagram is as follows:



Section 2: Contest Notes

- For certain system functionalities, if additional fields are required, please expand the fields with names starting with "New_" (e.g., New_MyColumn1), and the names of newly added columns must be in English.
- The solution file (filename) should be named **WML.sln**. Complete the system functions according to the requirements outlined in the "**Section 3: Design Requirements**". Deduct 1 point for incorrect filenames.
- The database linked by the application should be named **WML_ANM** (where "NM" represents the position number). Deduct 1 point for incorrect database names.
- When submitting SQL files, the filename must be **WML_ANM.bak** (where "NM" represents the position number). Deduct 1 point for incorrect SQL backup filenames or extensions. If the database cannot be restored, no points will be awarded for this competition.
- Create a **ReadMe.txt** file in the project directory (or application output directory) to explain the login credentials for users with different privileges. Deduct 2 points if the ReadMe.txt file is missing or named incorrectly.
- The user interface design should adhere to the design specification document:
 1. Deduct 1 point for each missing item: "System Name", "Alliance Identification (Logo)", and "Login/Logout Button."
 2. Deduct 2 points for not displaying the current user's name correctly (format: {First Name}{single space}{Last Name}).
 3. Deduct 1 point for not using the specified color palette.
 4. Deduct 1 point for not using the specified alliance identification.
 5. Deduct 1 point for not using the specified font.

Section 3: Design Requirements

The World Museum League (WML) ticket system management has two main functions: System Basic Requirements and Account Inquiry Functionality. The detailed descriptions of each function are as follows:

(1) System Basic Requirements

- Form Title

1. Title Content: 54th_World Museum League

2. The application icon must use the Logo file.



- Password Hashing Functionality

1. Hash the data in the Account.HashPassword field using the SHA256 algorithm, and overwrite the original field with the new value.

2. The code for performing the hashing operation should be stored in a separate class file.

- The file name must be "ProcessSHA256.cs" and stored in the project root directory.

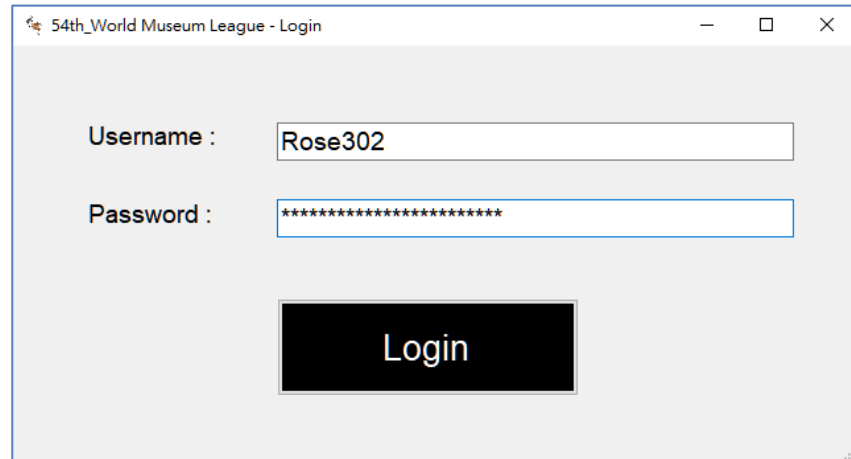
- The class name in the file should be the same as the file name (excluding the file extension).

3. Example code for hashing:

```
using (SHA256 sha256 = SHA256.Create())
{
    byte[] clearValue = Encoding.UTF8.GetBytes("[The text to be hashed.]");
    byte[] hashedValue = sha256.ComputeHash(clearValue);
    StringBuilder builder = new StringBuilder();
    for (int i = 0; i < hashedValue.Length; i++)
    {
        builder.Append(hashedValue[i].ToString("x2"));
    }
    string hashedString = builder.ToString();
}
```

- Login/logout function

1. Provide an administrator login interface where the account login method allows logging in via "Account" and "Password". The interface should be the following image, with the password input field not displaying plaintext:



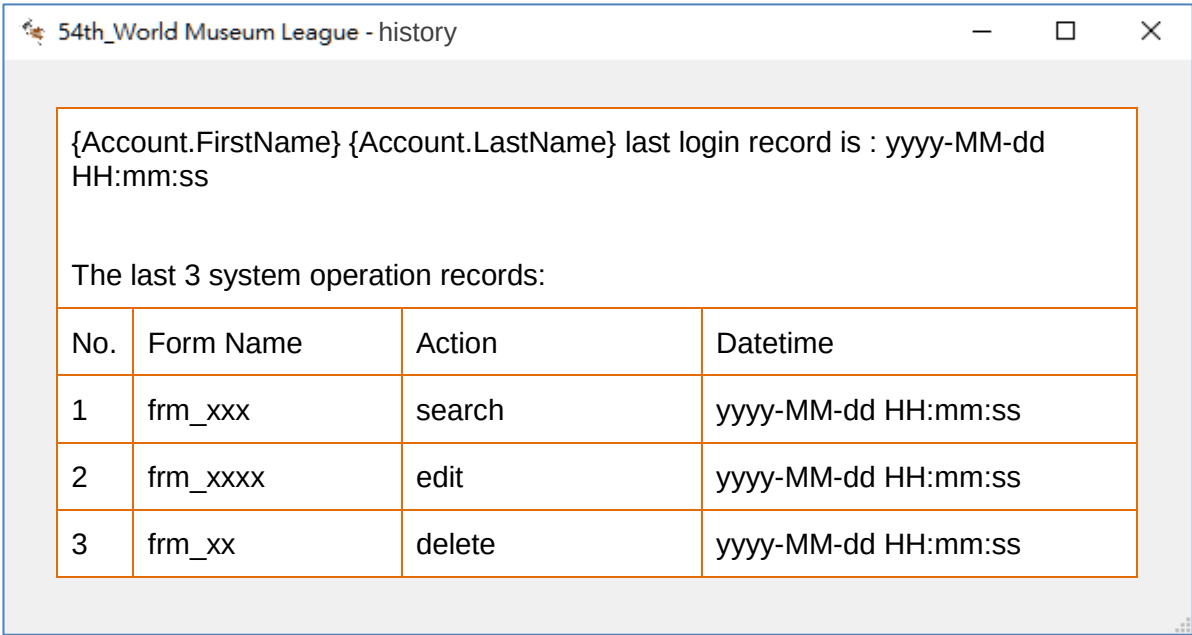
2. In case of incorrect account or password, display: "Incorrect account or password", and login should not proceed. (If SHA256 hash values are not used to verify passwords, deduct 3 points).
3. If the account data is correct but the account is disabled, display: "The account has been locked and cannot be logged in", and login should not proceed. The account status (Account.AccountStatus) is defined as follows:
 - 1 : Normal
 - 2 : Deactivate
4. After all checks are completed, if the logged-in user type is a regular visitor, display: "Visitors cannot use this management system"; if it is an administrator, they can log in to the system normally, and display: "Administrator {Account.FirstName} {Account.LastName}, welcome to use the World Museum League pass management system." The account type (Account.AccountType) is defined as follows:
 - 1 : Administrator
 - 2 : Visitor
5. After clicking the logout button, log out the currently logged-in account and prompt: "Logout Successfully !", then return to the login page.
6. In the logged-in state, if the program is closed under any circumstances, upon reopening and logging in with the same management account, it should return to the

last visited functionality of that management account (data does not need to be carried over).

(2) Show historical records function

- This account displays the date and time of the last login. (2 points)
- This account displays the most recent 3 system operation records, with correct content and format in the fields. (8 points, each error deducts 2 points)

1. Reference:



{Account.FirstName} {Account.LastName} last login record is : yyyy-MM-dd HH:mm:ss

The last 3 system operation records:

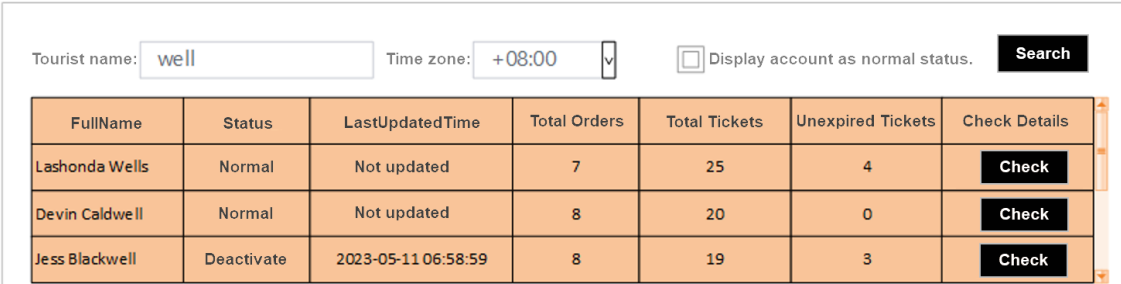
No.	Form Name	Action	Datetime
1	frm_xxx	search	yyyy-MM-dd HH:mm:ss
2	frm_xxxx	edit	yyyy-MM-dd HH:mm:ss
3	frm_xx	delete	yyyy-MM-dd HH:mm:ss

(3) Account Inquiry Functionality

- All administrators are able to access this feature, which is used to manage the data of "non-administrator" accounts.
- Upon entering this feature, all account data should be listed.
- Please provide filtering function:
 1. Tourist Name: Provide a single text box to search for tourist data by fuzzy matching Last Name or First Name.
 2. Display account as normal status: Checkbox.
 3. Ability to select Time zone: The time zone list should include options from - 12:00 to +14:00, with each hour as a unit.

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- After pressing Search, the following data is queried:
 1. FullName: Comprised of First Name and Last Name.
 2. Status: Displayed as "Normal" or "Deactivate".
 3. LastUpdatedTime: Displayed in the format "yyyy-MM-dd HH:mm:ss", shown in the selected time zone.
 4. Total Orders: The number of paid orders.
 5. Total Tickets: The total number of tickets in valid orders.
 6. Unexpired Tickets: The tickets that are still valid at the current system time.
 7. Check: Presented as a button, clicking on it opens detailed information.
 8. All columns except "Check Details" can be clicked on the header to sort in ascending or descending order.
- 9. Reference:



Tourist name: Time zone: ☐ Display account as normal status.

FullName	Status	LastUpdatedTime	Total Orders	Total Tickets	Unexpired Tickets	Check Details
Lashonda Wells	Normal	Not updated	7	25	4	Check
Devin Caldwell	Normal	Not updated	8	20	0	Check
Jess Blackwell	Deactivate	2023-05-11 06:58:59	8	19	3	Check

(4) Order Overview Functionality

- When clicking the "Check" button, display the account's detailed information on the same screen.
 1. Full Name: Display the combination of First Name and Last Name, along with the login account. Provide administrators with the option to deactivate or activate the account. When the account is deactivated or activated, update the content of the search table immediately.
 2. Order Number: Display the order number.
 3. Amount, Discount, Total Amount: Displayed to two decimal places with thousands separators.
 4. Paid Time, Create Time: Format as "yyyy-MM-dd HH:mm:ss", displayed in the selected time zone. Display "N/A" if not yet paid.

5. Number of Tickets: The number of tickets included in the order.
6. Except for "Order Number", all headers in the table can be clicked to sort in ascending or descending order.
7. The data in the Ticket Number column is clickable, leading to the detailed content of the order when clicked.
8. Reference:

Tourist name:
Time zone:
☐ Display account as normal status.

Full Name	Status	Data Update Time	Total Orders	Total Tickets	Unexpired Tickets	Check Details
Lashonda Wells	Normal	Not updated	7	25	4	Check
Devin Caldwell	Normal	Not updated	8	20	0	Check
Jess Blackwell	Deactivate	2023-05-11 06:58:59	8	19	3	Check

FullName: Devin Caldwell (Login ID: Lashonda3724)- [Deactivate](#)

Order Overview

OrderNumber	Amount	Discount	Total Amount	PaidTime	CreatedTime	NumberOf Tickets
JUKB532134148617875460658	310.00	1.05	308.95	2021-08-16 06:24:18	2021-08-16 05:42:23	3
RNFx833949834301365308900	310.00	0.00	310.00	2022-03-09 11:44:04	2022-03-09 10:56:48	3
GTGE726010058543687344829	912.50	242.40	670.10	2022-06-24 23:10:19	2022-06-24 22:44:15	4

(5) Ticket Overview Functionality

- When clicking on the OrderNumber data, display the ticket details on the same screen.
 1. Ticket Number: Display the order number of the ticket to view.
 2. Named: If the ticket is a named ticket, display the full name; if it is an anonymous ticket, display "Anonymous".
 3. Effective Time Start, Effective Time End: If the ticket has been activated, display the time in the format "yyyy-MM-dd HH:mm:ss", shown in the selected time zone; if not yet activated, display "--".
 4. Applied Age Type, Applied Duration Type: Display content according to the alliance rules.
 - Applied Age Type: 1=Child; 2=Teenager; 3=Adult; 4=Elder
 - Applied Duration Type: 1=Day; 2=Half Year; 3=Year

5. QR Code: The image file name used within the field is "QRCode.png". When clicked, please utilize the provided QRCodeSample project file and make modifications to generate a QR code for the Ticket Number with a minimum size of 200px * 200px. Apply the Frame.png image file and open a new window to display the result. Reference image:



6. Reference:

Tourist name:
Time zone:
☐ Display account as normal status.

Full Name	Status	Data Update Time	Total Orders	Total Tickets	Unexpired Tickets	Check Details
Lashonda Wells	Normal	Not updated	7	25	4	Check
Devin Caldwell	Normal	Not updated	8	20	0	Check
Jess Blackwell	Deactivate	2023-05-11 06:58:59	8	19	3	Check

Full Name: Devin Caldwell (Login ID: Lashonda3724)- Deactivate

Order Overview

OrderNumber	Amount	Discount	Total Amount	PaidTime	CreatedTime	NumberOf Tickets
<u>JUKB532134148617875460658</u>	310.00	1.05	308.95	2021-08-16 06:24:18	2021-08-16 05:42:23	3
<u>RNFX833949834301365308900</u>	310.00	0.00	310.00	2022-03-09 11:44:04	2022-03-09 10:56:48	3
<u>GTGE726010058543687344829</u>	912.50	242.40	670.10	2022-06-24 23:10:19	2022-06-24 22:44:15	4

Ticket Number

Ticket overview

Ticket Number	Named	EffectiveTimeStart	EffectiveTimeEnd	Applied AgeType	Applied Duration Type	QRCode
JUKB532134148617875460658-001	Floyd Bridges	2021-08-28 06:53:39	2021-08-29 06:53:39	Teenager	Day	
JUKB532134148617875460658-002	Anonymous	2021-09-11 14:41:39	2022-03-10 14:41:39	Adult	Half Year	
JUKB532134148617875460658-003	Anonymous	2021-09-04 08:40:22	2021-09-05 08:40:22	Adult	Day	

Section 4: Drawing System Analysis Diagram

- Please draw a Use Case Diagram for the "Account Lookup" feature of this system.
 1. Please create a Use Case Diagram and save it as a VISIO file named "UseCaseDiagram.VSDX" (1 pt)
 2. Are the roles (Actors) correct? (1 pt)
 3. Is the system boundary correct? (1 pt)
 4. Are the associations between actors and use cases reasonable? (2 pts)
 5. Are the descriptions of Include and Extend relationships correct? (2 pts)
 6. Does the Use Case Diagram design align with the data table design and is it complete? (3 pts)

Section 5: Presentation and Oral Report

- From the perspective of a system development engineer, explain the system development, including its functionalities, processes, and design principles.
 1. Create a slide presentation that includes a header, footer, topics, functions, processes, design philosophy, etc., adhering to the Style Guidelines (WML Style Guide.pdf). Ensure that each slide includes page number and at least two system images.
 2. The presentation time for the oral report is limited to 5 minutes.
 3. The oral report can be delivered in any language.

	Description	Mark	Value
(1)	<u>Contest Notes</u> The SQL filename must be WML_ANM.bak (where "NM" represents the position number). <u>If the database cannot be restored, no points will be awarded for this competition.</u> ■ Deduct 1 point for incorrect database names. ■ Deduct 1 point for incorrect SQL backup filenames or extensions. ■ Deduct 1 point for incorrect WML.sln solution filename. ■ Deduct 2 points if the ReadMe.txt file is missing or named incorrectly.	5	
(2)	<u>Design Requirements</u> ■ Deduct 1 point for each missing item: "System Name", "Alliance Identification (Logo) ", and "Login/Logout Button." (3 pts) ■ Deduct 2 points for not displaying the current user's name correctly (format: {First Name}{single space}{Last Name}). ■ Deduct 1 point for not using the specified color palette. ■ Deduct 1 point for not using the specified alliance identification. ■ Deduct 1 point for not using the specified font.	8	
(3)	<u>System Basic Requirements</u> 【Form Title】 ■ Deduct 1 point for not using the correct form title format and text. ■ Deduct 1 point for not using the correct Logo file.  【Password Hashing Functionality】 ■ Deduct 1 point for incorrect file name "ProcessSHA256.cs" and stored in the project root directory. ■ Deduct 1 point for incorrect class name. ■ Deduct 3 points for incorrect design method content. ■ Deduct 3 points for incorrect hash database password field.	10	

Description		Mark	Value
(4)	【 Login/logout function 】 ■ Deduct 1 point for incorrect the password input field not displaying plaintext. ■ Deduct 3 points for incorrect SHA256 hash values are not used to verify passwords. ■ Deduct 1 point for incorrect when incorrect account or password and display "Incorrect account or password" message. ■ Deduct 1 point for incorrect the account is disabled and display "The account has been locked" message. ■ Deduct 1 point for incorrect if the logged-in user type is a regular visitor, display "Visitors cannot use this management system" message. ■ Deduct 1 point for incorrect an administrator can log in to the system normally and display "Administrator {Account.FirstName} {Account.LastName}, welcome to use the World Museum League pass management system." message. ■ Deduct 1 point for incorrect log out the currently logged-in account and display "Logout Successfully !" message. ■ Deduct 1 point for not return to the login page when logout successful. ■ Deduct 3 points for incorrect In the logged-in state, program is closed under any circumstances, upon reopening and logging in with the same management account, it should return to the last visited functionality of that management account.	13	
(5)	【 Show historical records function 】 ■ This account displays the date and time of the last login. (2 points) ■ This account displays the most recent 3 system operation records, with correct content and format in the fields. (8 points, each error deducts 2 points)	10	

Description		Mark	Value
(6)	【Account Inquiry Functionality】 ■ Deduct 2 points for incorrect manage the data of "non-administrator" accounts, and all account data should be listed. ■ Deduct 1 point for incorrect filtering function. ■ Deduct 1 point for incorrect clicking the Check button and opening the detail information. ■ Deduct 2 points for incorrect number of records. ■ Deduct 2 points for incorrect field definition. ■ Deduct 1 point for incorrect field names are displayed. ■ Deduct 1 point for incorrect account Status filed is displayed. ■ Deduct 2 points for incorrect LastUpdatedTime filed is displayed and need to include time zone. ■ Deduct 3 points for incorrect Total Orders, Total Tickets, and Unexpired Tickets fields are displayed. ■ Deduct 1 point for incorrect CheckDetails button style. ■ Deduct 2 points for incorrect sorting function. 【Order Overview Functionality】 ■ Deduct 1 point for incorrect Full Name displayed. ■ Deduct 2 points for incorrect Logic for enabling or disabling accounts. ■ Deduct 1 point for incorrect number of records. ■ Deduct 2 points for incorrect field definition. ■ Deduct 4 points for incorrect PaidTime and CreatedTime fields are displayed and need to include time zone. ■ Deduct 2 points for incorrect sorting function. ■ Deduct 1 point for incorrect OrderNumber style.	50	

	Description	Mark	Value
	【 Ticket Overview Functionality 】 ■ Deduct 1 point for incorrect Ticket Number displayed. ■ Deduct 1 point for incorrect number of records. ■ Deduct 2 points for incorrect field definition. ■ Deduct 1 point for incorrect Named displayed. ■ Deduct 4 points for incorrect the query results and values are correct and the time is included in the time zone. ■ Deduct 2 points for incorrect QRCode image displayed and the image file name used within the field is "QRCode.png". ■ Deduct 8 points for incorrect (2 points for each of the following): (1) When clicked and using QRCodeSample project file and make modifications to generate a QR code. (2) The QRCode content is the Ticket Number field. (3) The QRCode of size is a minimum of 200px * 200px. (4) Apply the Frame.png image file and open a new window to display the result.		
(7)	<u>Programming Development Literacy</u> ■ Deduct 3 points for incorrect the system prevents crushing, preventing mismatches, and the auxiliary message prompts are well designed. ■ Deduct 3 points for incorrect the code contains comments. ■ Deduct 3 points for incorrect the code naming rules are meaningful and are not system defaults.	9	

Description		Mark	Value
(8)	<u>Drawing System Analysis Diagram</u> ■ Deduct 1 point for incorrect file name. ■ Deduct 1 point for incorrect the roles (Actors). ■ Deduct 1 point for incorrect the system boundary. ■ Deduct 2 points for incorrect the associations between actors and use cases reasonable. ■ Deduct 2 points for incorrect the descriptions of Include and Extend relationships. ■ Deduct 3 points for incorrect the diagram is complete.	10	
(9)	<u>Presentation</u> ■ Deduct 2 points for incorrect a slide presentation that includes a header, footer, topics, functions, processes, design philosophy. ■ Deduct 1 point for incorrect adhering to the Style Guidelines. ■ Deduct 1 point for incorrect each slide includes page number. ■ Deduct 1 point for incorrect at least two system images.	5	
(10)	<u>Oral briefing</u> ■ Deduct 1~3 points for the overall performance of the speech presentation. ■ Deduct 1~3 points for the charm of speech. ■ Deduct 1~3 points for the completeness of the speech. ■ Deduct 1 point for incorrect time control of speeches.	10	
Total		130	